Power Distribution: System Design Description

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Classification	INTERNAL
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1. System overview

Power distribution carries reactor and solar output to the zone substations over superconducting bus conduits, with backup generator banks and storage battery strings for life-critical loads.

The system is led by Reactor & Power Systems. It is delivered through 6 work packages with a combined budget of 70,230.45 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
PD-MBC	Main bus conduit	SHV-PD-SPEC-0101	SHV-PD-DWG-0201	SHV-PD-BOM-0301
PD-SUB	Zone substation	SHV-PD-SPEC-0102	SHV-PD-DWG-0202	SHV-PD-BOM-0302
PD-BKG	Backup generator bank	SHV-PD-SPEC-0103	SHV-PD-DWG-0203	SHV-PD-BOM-0303
PD-SBS	Storage battery string	SHV-PD-SPEC-0104	SHV-PD-DWG-0204	SHV-PD-BOM-0304

3. Architecture

3.1 Main bus conduit

The main bus conduit is specified in SHV-PD-SPEC-0101. Its governing parameter is the conduit rating, set at 10.3 % reactor. The responsible engineer is Tarn Delane.

3.2 Zone substation

The zone substation is specified in SHV-PD-SPEC-0102. Its governing parameter is the substation rating, set at 2.5 TW. The responsible engineer is Delia Bailin.

3.3 Backup generator bank

The backup generator bank is specified in SHV-PD-SPEC-0103. Its governing parameter is the bank output, set at 1.35 TW. The responsible engineer is Fallon Darrow.

3.4 Storage battery string

The storage battery string is specified in SHV-PD-SPEC-0104. Its governing parameter is the storage capacity, set at 101 GWh. The responsible engineer is Venn Nash.

4. Interfaces

Document	Title
SHV-ICD-HR-PD	Interface Control Document: Hypermatter Reactor / Power Distribution
SHV-ICD-SA-PD	Interface Control Document: Solar Collector Array / Power Distribution

Document	Title
SHV-ICD-PD-LS	Interface Control Document: Power Distribution / Life Support & Atmospherics
SHV-ICD-PD-HB	Interface Control Document: Power Distribution / Docking & Hangar Bays
SHV-ICD-PD-SY	Interface Control Document: Power Distribution / Shipyard Slipways
SHV-ICD-PD-PR	Interface Control Document: Power Distribution / Propulsion & Station-Keeping
SHV-ICD-PD-AG	Interface Control Document: Power Distribution / Hydroponics & Food Production
SHV-ICD-PD-TR	Interface Control Document: Power Distribution / Internal Transit
SHV-ICD-PD-RL	Interface Control Document: Power Distribution / Research Laboratories

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-PD-01	Power Distribution: Design & analysis	Tamsin Quarrel	IN_PROGRESS	83.1%
WP-SHV-PD-02	Power Distribution: Fabricate & install main bus conduit	Hollis Fenwick	COMPLETE	100.0%
WP-SHV-PD-03	Power Distribution: Fabricate & install zone substation	Edda Lancer	COMPLETE	100.0%
WP-SHV-PD-04	Power Distribution: Fabricate & install backup generator bank	Ulric Kirwan	IN_PROGRESS	95.4%
WP-SHV-PD-05	Power Distribution: Fabricate & install storage battery string	Nils Errant	IN_PROGRESS	79.6%
WP-SHV-PD-06	Power Distribution: Integration & test	Hollis Fenwick	IN_PROGRESS	65.1%

6. Risks

- Supply of main bus conduit assemblies depends on a single supplier.
- Supply of zone substation assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.